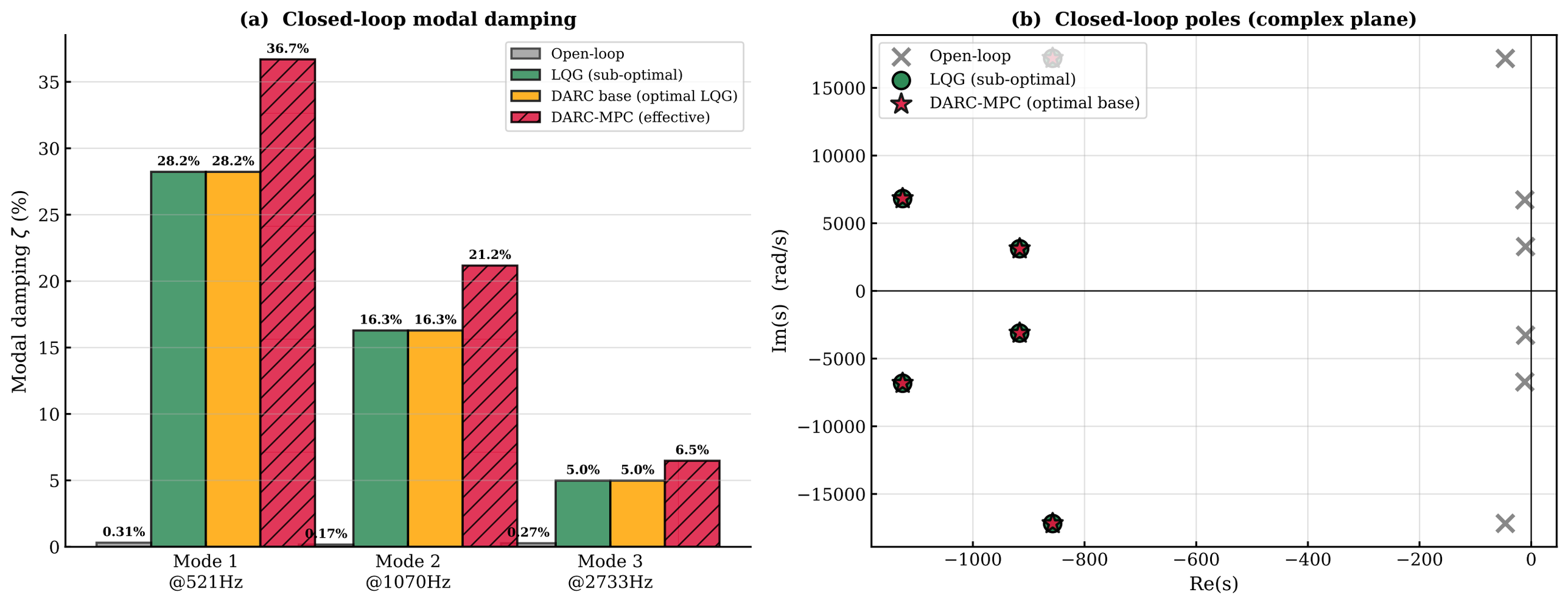
(a) Closed-loop modal damping
Open-loop
LQG (sub-optimal)
DARC base (optimal LQG)
DARC-MPC (effective)
Modal damping ζ (%)
0.31%
28.2%
28.2%
36.7%
0.17%
16.3%
16.3%
21.2%
0.27%
5.0%
5.0%
6.5%
Mode 1
@521Hz
Mode 2
@1070Hz
Mode 3
@2733Hz
(b) Closed-loop poles (complex plane)
Open-loop
LQG (sub-optimal)
DARC-MPC (optimal base)
Im(s) (rad/s)
Re(s)